

TOPOLOGY-OPTIMIZED BRACKET REDESIGN FOR WEIGHT REDUCTION

An Engineering Design & Analysis Report

Component Name - GE Jet Engine Bracket

Prepared by: Krishna Ganeshram

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Tools used: SolidWorks, CATIA, Ansys Mechanical(25R2), Ansys Topology Optimization

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Abstract

Material: Titanium Ti-6Al-4V, yield strength 903 MPa (131 ksi), per the official GE Jet Engine Bracket Challenge specification [2]. Elastic and ultimate strength properties (Section 3.1) are taken from a matching Ti-6Al-4V datasheet [1]

Service Temperature: 23.9°C (75°F)

Min. wall thickness: 10mm

Function: Structural mounting bracket that interfaces the jet engine to the aircraft pylon/strut via a clevis pin and bolted mounting points.

Constraints (fixed geometry): the load is applied at a 3/4" diameter pin at interface 1, Interfaces 2–5: 0.375-24 AS3239-26 machine bolts (fixed supports, infinitely stiff).

Load Cases:

Case	Type	Magnitude	Direction
LC1	Static linear	35,586 N	Vertical
LC2	Static linear	37,810 N	Horizontal, out
LC3	Static linear	42,258 N	42° from vertical
LC4	Static torsional	564,924 N·mm	Horizontal, at pin centerline intersection

The bracket was fixed with four bolts, and the pin diameter was 19.05 mm.

The original GE baseline STEP file is no longer publicly available; a baseline design was therefore constructed to satisfy the published envelope, interface locations, and load/boundary conditions from the official challenge specification.

1. Introduction & Problem Statement

1.1 Component Context

This component is a structural mounting bracket from the GE Jet Engine Bracket Challenge (GrabCAD, 2013): it clevis-pins to the jet engine at Interface 1 and bolts to the aircraft pylon/strut at Interfaces 2–5, transmitting the engine's static and inertial loads into the airframe. Because it is carried aloft on every flight as non-optimized structural mass, reducing its weight directly reduces fleet-level fuel burn without changing the engine's performance envelope — the original motivation for GE's public challenge to crowdsource a lighter redesign.

1.2 Design Requirements

- **Function:** Transmits static loads from the jet engine into the aircraft pylon/strut. The engine-side load is applied through a 19.05mm (0.75") clevis pin at Interface 1; the airframe-side is a rigid bolted mount at Interfaces 2–5. The bracket itself does not move in service.

- Material: Titanium Ti-6Al-4V (see Abstract for the specific yield strength and condition used). Chosen for its high specific strength and established aerospace qualification, consistent with the official challenge material assumption.
- Load case(s): LC1: 35,586 N vertical; LC2: 37,810 N horizontal (out); LC3: 42,258 N at 42° from vertical; LC4: 564,924 N·mm torsional about the pin axis. All four are applied individually (not combined) at the Interface 1 clevis pin bore.
- Boundary conditions / constraints: Interfaces 2–5 (four 0.375-24 AS3239-26 machine bolt holes in the Base) are fixed supports, treated as infinitely stiff. The 19.05mm pin at Interface 1 is likewise treated as infinitely stiff and is not itself part of the optimized mass.
- Target factor of safety: $FOS \geq 1.5$ on yield, treating LC1–LC4 as limit loads — consistent with the official challenge’s Phase I performance criteria, under which a design that meets this target is then pushed to an increased ultimate-capability load in a separate check.

1.3 Objective

Reduce the structural mass of the titanium jet engine mounting bracket while maintaining $FOS \geq 1.5$ on yield under each of LC1–LC4 individually, and produce a manufacturable redesign of the topology optimizer's raw output that preserves the fixed clevis-pin and bolted-mount interfaces.

2. Theoretical Background

This section summarizes the analytical basis for the finite element and topology optimization methods used in this study, establishing the terminology and assumptions referenced in later sections.

2.1 Finite Element Analysis Fundamentals

Finite element analysis (FEA) approximates a continuous structure by discretizing it into a finite number of elements connected at nodes. Each element's local stiffness matrix k relates nodal displacements to nodal forces; these are assembled into a global stiffness matrix K covering the whole mesh. For a linear-elastic static analysis, equilibrium reduces to the matrix equation $KU = F$, which is solved for the global nodal displacement vector U given the applied load vector F and the prescribed (fixed) degrees of freedom.

This study assumes linear-elastic material behavior (stress proportional to strain per Hooke's law) and small-displacement theory, so geometric nonlinearity is neglected. From the solved nodal displacements, elemental strains are recovered via strain-displacement relations, and stresses follow from the constitutive relation $\sigma = D\epsilon$, where D is the material stiffness matrix built from Young's modulus E and Poisson's ratio ν .

Because the applied loads and constraints in this problem act in multiple directions simultaneously, results are compared using the von Mises equivalent stress — a scalar measure derived from the full stress tensor that collapses a multiaxial stress state into a single value comparable against a uniaxial yield strength, per the distortion-energy (von Mises) failure theory.

Discretization introduces approximation error that decreases as element size decreases. The mesh sensitivity and quality study (Section 4.2) evaluates how the reported peak stress changes with mesh refinement and identifies regions where the result is sensitive to mesh resolution.

2.2 Topology Optimization Theory

Topology optimization used in this study follows the density-based SIMP method (Solid Isotropic Material with Penalization). Each element in the design domain is assigned a pseudo-density $\rho_e \in [\rho_{\min}, 1]$, where $\rho_{\min}$ is a small nonzero floor used to avoid a singular stiffness matrix at zero density. An element's stiffness is scaled by ρ_e^p , where p is a penalization exponent (typically 3), which discourages intermediate ("grey") densities and drives the solution toward a clear solid/void distribution.

The optimizer minimizes structural compliance $c(\rho) = \mathbf{U}^T \mathbf{K}(\rho) \mathbf{U}$ (equivalently, maximizes stiffness) subject to a target volume fraction and the equilibrium constraint $\mathbf{K}(\rho)\mathbf{U} = \mathbf{F}$. A sensitivity (adjoint) analysis computes the gradient of compliance with respect to each element's density, which an iterative update scheme (e.g., Optimality Criteria or Method of Moving Asymptotes) uses to redistribute material toward the load-carrying paths. Density or sensitivity filtering is applied to prevent checkerboard patterns and to control minimum feature size in the result.

The optimizer's raw output is a density field — typically visualized as an organic, non-manufacturable iso-surface mesh (Section 5.2) — representing a theoretical lower bound on mass for the given stiffness/stress target. It is not itself a finished part; translating it into a producible geometry is the purpose of Section 6, Redesign for Manufacturability.

2.3 Factor of Safety and Failure Criteria

Factor of safety (FOS) is defined here as $\text{FOS} = \sigma_{\text{yield}} / \sigma_{\text{applied}}$, using the von Mises equivalent stress for σ_{applied} , appropriate for a ductile metal under static loading. This is a yield-based (not ultimate-based) convention: it indicates the margin against the onset of permanent deformation, not fracture.

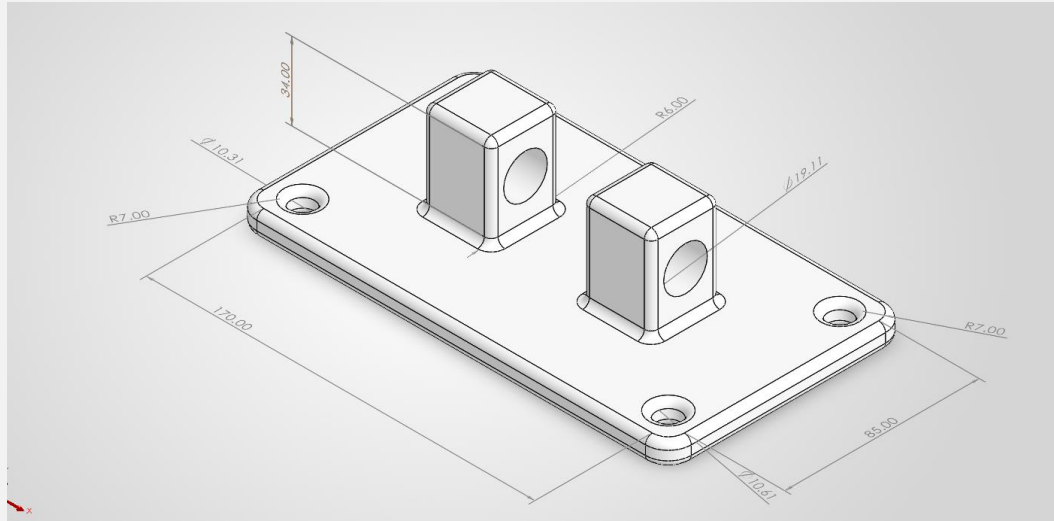
In this study, LC1–LC4 are treated as limit loads: the maximum static loads the bracket is expected to see in service. A target $\text{FOS} \geq 1.5$ on yield is applied against these limit loads, matching the official challenge's Phase I performance criteria. Designs meeting this target would, per the official rules, subsequently be pushed to an increased load to demonstrate ultimate capability — a separate check not performed in this study.

3. Baseline Design

This section documents the V2 baseline, which supersedes the V1 geometry used earlier in development. V2 keeps the same overall envelope ($190 \times 105 \times 10\text{mm}$ base plate), the same interface locations, and the same load application points and boundary conditions as V1. The bracket is modeled in SolidWorks as a single continuous solid (Base and the two boss towers/clevis lugs as one body, not a bonded assembly) to reflect that this is one machined/cast part in service. The two boss towers are symmetric about the base plate's width centerline and each carries a $\varnothing 19.11\text{mm}$ bore for the clevis pin. All fillets, hole positions, and interface dimensions are parametrically driven, so the geometry can be reproduced or resized directly from the dimensions given in Figures 1 and 2.

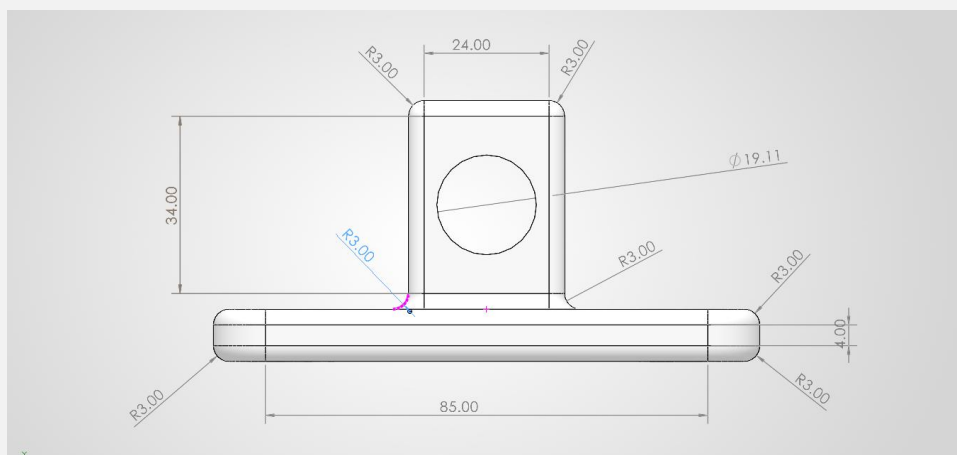
V1→V2 change and rationale: V1 used a sharp, essentially unfilleted interior corner where each clevis lug met the base. Mesh refinement at that corner produced a classic non-converging peak stress (see Section 4.2), which is the

signature of a stress singularity rather than a normal discretization error. V2's principal geometric change is a systematic 3mm (R3) fillet applied at every boss-to-base and boss-to-boss transition — adopted here as the V2 baseline fillet radius — in place of V1's sharp/minimally-filleted corners. The goal was to remove the singularity by giving the mesh a resolvable geometric feature to converge on; Section 4.2 documents how much of that goal was actually achieved.



Isometric view, SolidWorks V2 baseline model — 3mm (R3) fillets added at every boss-to-base and boss corner transition; Ø19.11mm pin bores on the 190×105×10mm mounting flange (Base)

Figure 1. V2 baseline bracket design (fillets added), isometric view.



Front section view — 3mm (R3) fillets applied throughout (boss-to-base transition, boss top corners, base plate edges); boss cross-section 24mm wide × 34mm tall with Ø19.11mm pin bore, 105mm base width, 10mm base thickness

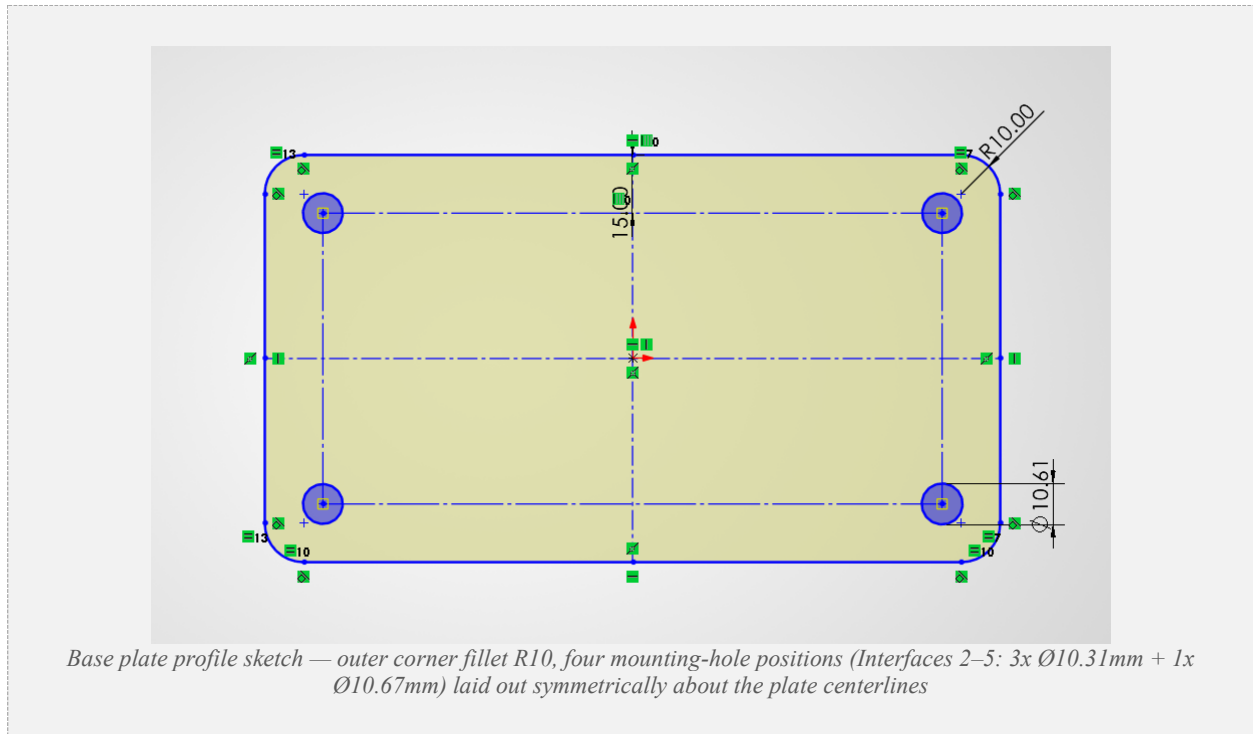
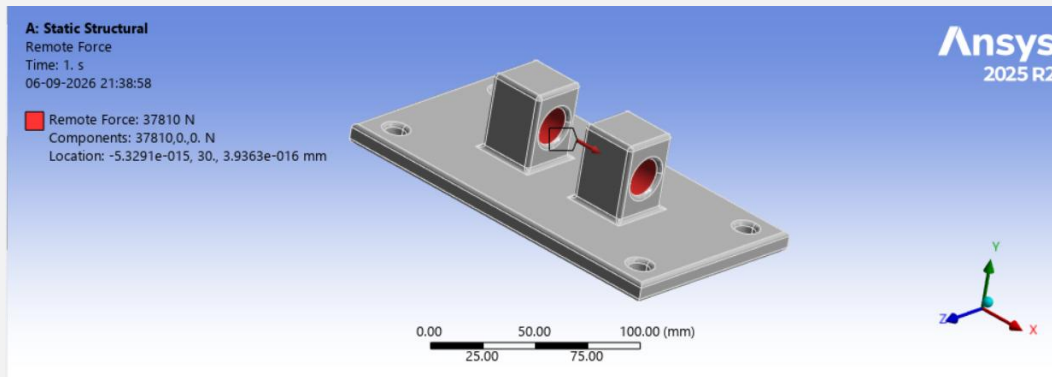


Figure 2. V2 baseline bracket, dimensioned views (envelope $190 \times 105 \times 10$ mm; plate corner fillet R7–R10; boss fillets R3/3mm typ.).

3.1 Material Properties

Property	Symbol	Value	Units	Source
Young's Modulus	E	113.8	GPa	MatWeb, Ti-6Al-4V (annealed) [1]
Yield Strength	σ_y	903	MPa	Per Abstract / challenge material assumption [2]
Ultimate Tensile Strength	σ_u	950	MPa	MatWeb, Ti-6Al-4V (annealed) [1]
Density	ρ	4430	kg/m ³	MatWeb, Ti-6Al-4V [1]
Poisson's Ratio	ν	0.342	—	MatWeb, Ti-6Al-4V [1]

3.2 Load Cases & Boundary Conditions



Remote Force applied to the clevis bore inner faces (LC2 shown as a representative example of the method; the same setup pattern is used for LC1, LC3, and LC4).

Figure 3. Applied loads and boundary conditions.

Case	Description	Magnitude	Direction	Application Location
LC1	Static linear, vertical limit load	35,586 N	Vertical (+Y)	Interface 1, clevis pin bore (remote force on bore inner faces)
LC2	Static linear, horizontal (out) limit load	37,810 N	Horizontal, out (+Z)	Interface 1, clevis pin bore
LC3	Static linear, 42° from vertical — resolved into orthogonal components (see note below the table)	42,258 N (resultant)	42° from vertical, toward +Z (out)	Interface 1, clevis pin bore
LC4	Static torsional load about the pin axis	564,924 N·mm	Horizontal, moment about pin centerline	Interface 1, pin centerline, midpoint between clevis arms

Note on LC3 implementation: Ansys Mechanical's Remote Force does not accept a load specified directly at an angle — only Cartesian (X/Y/Z) components. LC3's 42,258 N resultant, applied 42° from vertical, was therefore resolved into two orthogonal components before being entered as a single Remote Force: a vertical component = $42,258 \text{ N} \times \cos(42^\circ) = 31,404 \text{ N}$, and a horizontal (out) component = $42,258 \text{ N} \times \sin(42^\circ) = 28,276 \text{ N}$.

4. Baseline Static Structural FEA

4.1 Mesh & Solver Setup

- Element type: Quadratic tetrahedral (program-controlled, Ansys default solid element for this geometry)
- Global mesh size (V2, 3mm fillet baseline): 8 mm (coarse) / 4 mm (medium) / 2 mm (fine), evaluated per Section 4.2; local Face Sizing (0.5 mm) with a 1.5 mm global size subsequently applied once mesh-quality issues were identified at the boss-to-boss inner corner
- Total element count across the V2 mesh sensitivity sweep (Section 4.2): 5,614 (coarse) / 34,329 (medium) / 209,369 (fine) / 473,537 (Face Sizing — 0.5 mm local at the boss corners, 1.5 mm global, 1.2 growth rate). The 473,537-element Face Sizing mesh is the final adopted mesh, used for all reported LC1 baseline results in Section 4.3
- Solver: Ansys Mechanical 2025 R2, static structural, linear elastic — same solver configuration used for the V1 baseline. The Direct (Sparse) solver was used for all V2 mesh levels reported in Sections 4.2 and 4.3; cross-solver verification (Direct vs. Iterative) was not repeated for V2 and is noted as an open item in Section 9.

4.2 Mesh Sensitivity and Quality Study

V1→V2 background: V1 used a sharp, essentially unfilleted interior corner between each clevis lug and the base. Mesh refinement at that corner produced a non-converging peak stress (138.72 → 180.5 → 285.9 MPa across three refinement levels) — the signature of a stress singularity. A 1 mm fillet reduced but did not remove the sensitivity (180 → 240 → 320 MPa, +33%/step) because the global mesh could not resolve the small radius. Enlarging the fillet to 2 mm with a locally targeted Face Sizing control produced a practical, quality-controlled V1 baseline (471.64 MPa) that was still not shown to be fully mesh-independent (0.5 mm refinement pushed it to 623 MPa); full V1 details are reproduced below Figure 3b for reference. V2 was created to test whether a larger, systematically-applied fillet would remove this sensitivity: 3 mm (R3) fillets were added at every boss-to-base and boss-to-boss transition, and 3 mm is adopted here as the V2 baseline fillet radius. The table and findings below document the resulting V2 mesh sensitivity study.

Mesh Level	Size (mm)	Elements	Avg. Skewness	Max Skewness	Max Stress (MPa)	Max Deform. (mm)	FOS	% Change (Stress)
Coarse	8	5,614	0.350	0.970	295.09	0.1651	3.06	—
Medium	4	34,329	0.2618	0.9658	239.55	0.1708	3.77	−18.8%
Fine	2	209,369	0.2493	0.900	317.20	0.1723	2.85	+32.4%
Face Sizing (0.5/1.5)	—	473,537	0.254	0.9123	348.62	0.1734	2.59	+9.9%

Skewness, deformation, and FOS are reported alongside stress for completeness. FOS values here use the same yield ÷ peak-stress convention as Section 4.3 (903 MPa ÷ peak stress), so they track stress inversely and add no new information on their own — but the skewness columns do: average skewness stays in a reasonable 0.25–0.35 range at every level, while max skewness sits above 0.9 at all four mesh levels, including the final Face Sizing mesh. That persistently poor worst-case skewness, unchanged by refinement, is consistent with the localized meshing difficulty at the boss-to-boss corner discussed below.

Discretization Sensitivity: Coarse-to-Medium refinement produced an 18.8% decrease in peak stress (295.09 → 239.55 MPa) — opposite in direction from V1’s coarse-to-medium jump, and superficially suggestive of convergence.

Non-Monotonic Behavior: Medium-to-Fine refinement reversed this trend, jumping 32.4% to 317.2 MPa; the Face Sizing mesh rose a further 9.9% to 348.62 MPa. Despite the larger, systematically-applied fillet, V2’s peak stress still does not converge monotonically with mesh refinement.

Diligence performed: A SolidWorks geometry check flagged apparent discontinuities (1 edge, 2 vertices) at the inner boss-to-boss corner, but the reported gap was 5×10^{-6} mm (5 microns) — roughly 100,000× smaller than even the finest local element (0.5 mm) — consistent with normal B-rep floating-point tolerance rather than a real modeling defect. Repeating the same four-level sweep with the fillet enlarged to 4 mm produced the same dip-then-rise pattern, shifted slightly lower ($\approx 280 / 260 / 310 / 323$ MPa), so the behavior is not simply a function of fillet size. A Hex Dominant meshing attempt at this geometry failed to complete after four separate attempts (no crash — the mesher itself did not converge on a hex/tet-transition mesh in a reasonable time), consistent with — though not proof of — a locally difficult geometric feature at the boss-to-boss corner, independent of meshing algorithm.

Interpretation: This evidence supports treating the 3-way boss-to-boss corner as a locally mesh-sensitive feature that neither the geometry check nor two fillet radii were able to fully resolve. It does not, on its own, rule out a true (if very local) stress concentration at this feature — a clean CAD gap-check and a non-crashing-but-non-converging hex attempt are consistent with several explanations, not only “purely a meshing artifact.” Following the same practical-baseline approach taken for V1, the Face Sizing mesh (473,537 elements, 3 mm fillet) is reported as the V2 baseline result in Section 4.3, with this limitation carried forward into Discussion (Section 9) rather than treated as resolved.

For reference, V1’s final quality-controlled mesh and skewness distribution are reproduced below.

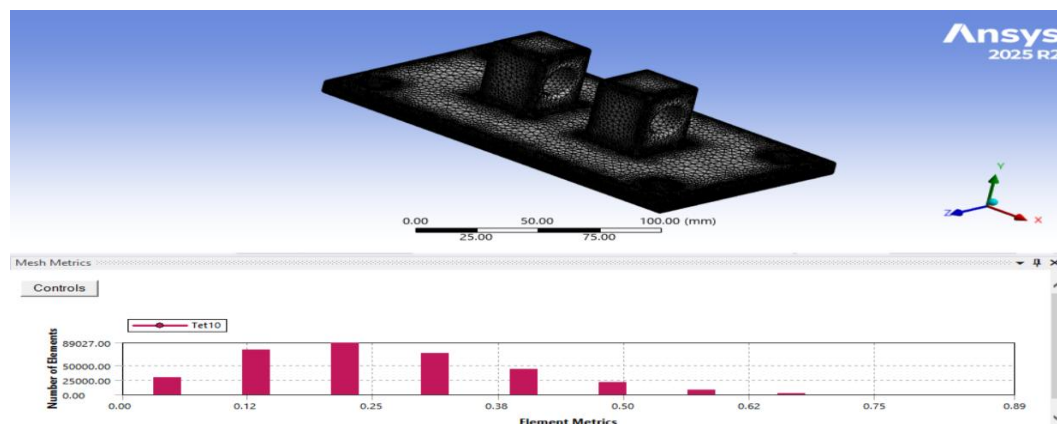
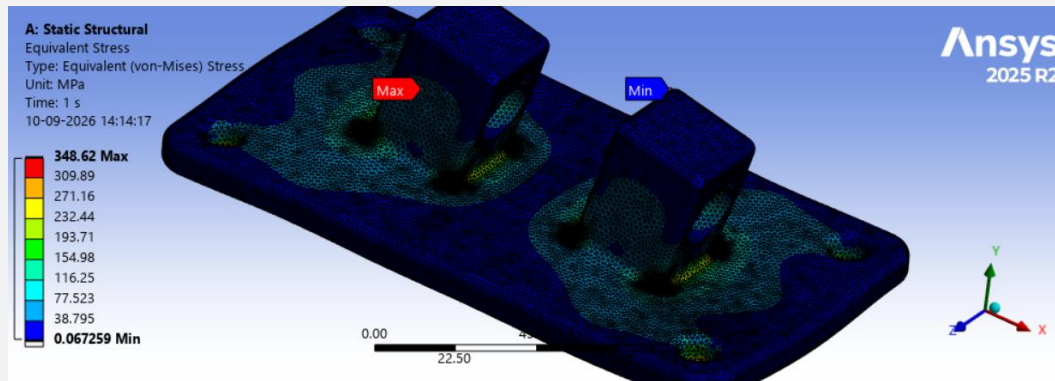


Figure 3b (V1 reference). V1’s final quality-controlled mesh (343,762 elements, 2 mm fillet) and skewness distribution — average 0.257, isolated outliers at the fillet corner not exceeding 0.93. Shown here for context on the meshing difficulty that motivated the V2 redesign; V2’s own mesh-level statistics are given in the table above.

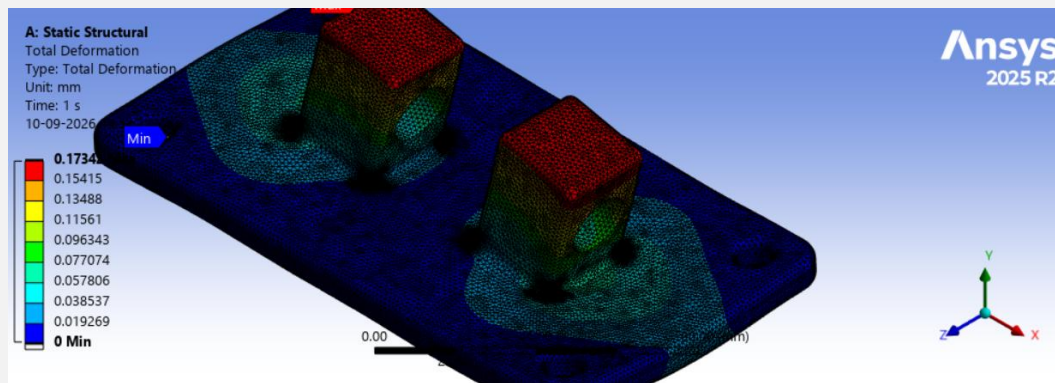
4.3 Baseline Results

The results below are for LC1 (35,586 N vertical) on the V2 final adopted mesh (473,537 elements, 3 mm fillet, Face Sizing — described in Section 4.2). FOS is computed directly as yield strength ÷ peak stress (903 MPa ÷ 348.62 MPa).



Von Mises stress contour plot — V2 baseline (3mm fillet, Face Sizing mesh)

Figure 4. Von Mises stress distribution, V2 baseline design, LC2 (37,810 N horizontal).



Total deformation contour plot — V2 baseline

Figure 5. Total deformation, V2 baseline design, LC2 (37,810 N horizontal).

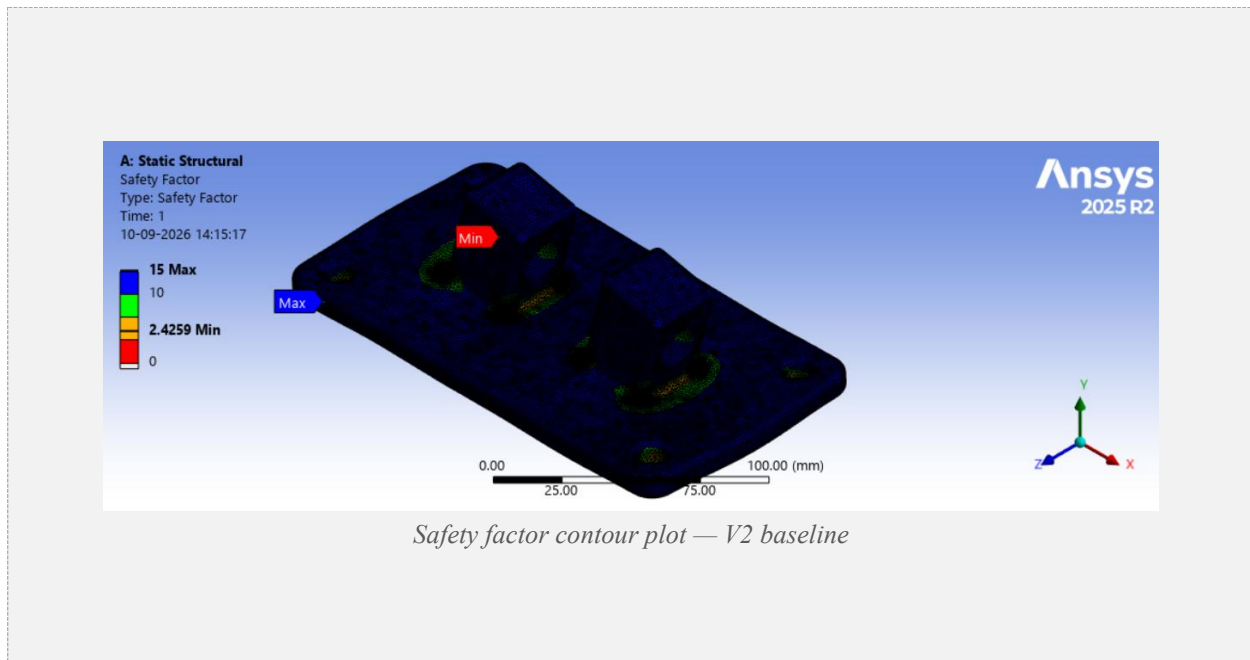


Figure 6. Factor of safety distribution, V2 baseline design, LC1 (35,586 N vertical).

Note: The plot above shows a minimum of 2.4259, computed by Ansys’s built-in Safety Factor tool using the yield strength entered in Engineering Data (847 MPa). This report instead states Minimum FOS as 2.59, computed directly as $903 \text{ MPa} \div 348.62 \text{ MPa}$, using the 903 MPa yield strength specified in this report’s Abstract and Material Properties table (Section 3.1). The two values differ only because of this yield-strength discrepancy between the Ansys model and the report; the underlying peak stress (348.62 MPa) is the same in both.

Metric	Value	Location
Max von Mises Stress	348.62 MPa	Boss-to-boss inner corner (between the two boss towers, loaded boss side)
Max Deflection	0.1734 mm	Top face of the unloaded boss tower
Minimum FOS	2.59	Same location as max stress (boss-to-boss inner corner)
Mass (baseline)	Mass = 1.094 kg	—

Peak stress concentrates at the inner-facing edge between the two boss towers — the 3-way corner intersection where each boss’s 3mm fillet meets the base and its neighboring boss — governing at 348.62 MPa. Geometrically, this corner sits directly in the load path: the vertical force applied at the pin bore must transfer through the boss wall, around the fillet, and into the base flange, concentrating bending and shear stress at the tightest curvature and geometrically most complex feature on the part. As documented in Section 4.2, this same corner proved the hardest region to mesh cleanly even after the fillet was enlarged from V1’s 1–2 mm to V2’s 3 mm baseline and re-tested at 4 mm, so this value should be read as a practical engineering estimate at a controlled mesh quality, not a demonstrated mesh-independent result.

5. Topology Optimization

5.1 Setup

- Design region: (to be defined once the optimization domain is set up)
- Exclusion / non-design regions: bolt holes, the clevis bore, and mating faces are expected to be excluded, since these are fixed functional interfaces that cannot change shape
- Objective: minimize mass (to be confirmed against the solver's actual objective function once configured)
- Constraints: [not yet run — replace with the exact constraint(s) used, e.g. maximum stress limit or retained volume fraction, once the optimization is configured in Ansys]

5.2 Raw Optimizer Output



Figure 7. Raw topology-optimized shape prior to redesign.

5.3 Interpretation

(To be completed once the optimizer has run: this should assess whether the resulting load path makes physical sense given LC1–LC4 — for example, whether material concentrates along the expected principal stress trajectory between the clevis bore and the mounting bolts, or whether it removes material somewhere unexpected that would need further investigation.)

6. Redesign for Manufacturability

Raw topology optimizer output is not itself a manufacturable part — it is an organic density field that must be reinterpreted into producible CAD geometry. This section documents that reinterpretation explicitly: the raw optimizer mesh is shown alongside the redesigned CAD, and every simplification decision (which features were smoothed, retained, or added back for manufacturability) is stated and justified below, so the redesign’s connection to the optimizer’s actual load-path recommendation remains traceable.

[INSERT IMAGE]

Raw topology mesh vs. redesigned CAD, side by side

Figure 8. Raw topology-optimized geometry (left) vs. manufacturable redesign (right).

6.1 Simplifications Made

- (To be completed once the redesign is done — expected to include smoothing the optimizer’s organic surfaces into swept/lofted solid features.)
- (To be completed: fillet sizes added at load-path transitions for stress relief, informed by the sensitivity already observed at the baseline’s own fillet corner in Section 4.2.)
- (To be completed: draft angle and/or minimum wall thickness enforced, depending on the manufacturing route chosen in Section 6.2.)
- (To be completed: note whether symmetry was retained to simplify fixturing or inspection.)

6.2 Manufacturing Route

(To be completed once a manufacturing route — CNC machining, casting, or metal additive manufacturing — is selected and justified against the redesigned geometry’s features.)

7. Validation FEA — Optimized Design

Same load cases, mesh strategy, and solver settings as the baseline (Section 4), applied to the redesigned geometry, for a fair before/after comparison.

[INSERT IMAGE]

Von Mises stress contour plot — optimized design

Figure 9. Von Mises stress distribution, optimized design (to be completed once the redesign is validated).

[INSERT IMAGE]

Total deformation contour plot — optimized design

Figure 10. Total deformation, optimized design (to be completed once the redesign is validated).

[INSERT IMAGE]

Factor of safety contour plot — optimized design

Figure 11. Factor of safety distribution, optimized design.

8. Results & Comparison

Metric	Baseline	Optimized	% Change
Mass	1.094 kg	[] kg	[]%
Max von Mises Stress	348.62 MPa	[] MPa	[]%
Minimum FOS	2.59	[]	[]%
Max Deflection	0.1734 mm	[] mm	[]%
Manufacturing Route	[]	[]	—

[3-5 sentences discussing trade-offs. E.g. did stiffness drop even though strength FOS held? Note it, don't hide it — reviewers respect that.]

9. Discussion & Limitations

This section states plainly what this study did not cover, alongside what it did. The mesh sensitivity finding documented in Section 4.2 (348.62 MPa as a practical, non-fully-converged V2 baseline) is carried forward here rather than treated as resolved, along with the other scope limitations below.

- Load cases: [e.g. only static limit-load cases were analyzed; fatigue/vibration not assessed]
- Validation: [no physical/experimental testing performed; results are simulation-only]
- Solver limitations: [e.g. Ansys Student element-count cap, linear-elastic assumption]
- Solver cross-validation: Direct vs. Iterative solver cross-validation was performed for the V1 baseline (both returned an identical peak stress), but was not repeated for the V2 baseline. The V2 result of 348.62 MPa has therefore not been independently confirmed as solver-independent.
- Manufacturing: [redesigned part not yet quoted/DFM-reviewed by a supplier]

10. Conclusion & Future Work

[Summarize the headline result in one or two sentences. Then list 2-3 concrete next steps, e.g.: add a fatigue load case, validate against physical prototype, explore alternative materials, extend to a multi-load-case optimization.]

References

- [1] MatWeb, "Titanium Ti-6Al-4V (Grade 5), Annealed," ASM Aerospace Specification Metals, Inc. <http://asm.matweb.com/search/SpecificMaterial.asp?bassnum=MTP641> (accessed September 2026). Source for Young's Modulus, Ultimate Tensile Strength, Density, and Poisson's Ratio.
- [2] General Electric / GrabCAD, "GE Jet Engine Bracket Challenge," official design specifications and rules, 2013. <https://grabcad.com/challenges/ge-jet-engine-bracket-challenge> and <https://blog.grabcad.com/ge-terms-of-service/>. Source for interface geometry, load cases LC1–LC4, the 131 ksi (~903 MPa) yield strength assumption, and the AMS4911 material-properties guidance.
- [3] Ansys, Inc., Ansys Mechanical User's Guide, Release 2025 R2, 2025. <https://ansyshelp.ansys.com>
- [4] Bendsoe, M.P. & Sigmund, O., Topology Optimization: Theory, Methods, and Applications, 2nd ed., Springer, 2003. Source for the SIMP method and topology optimization theory in Section 2.2.